

Response to Reviewers

Manuscript: A Kirkpatrick-based Framework for Integrating Safety and Sustainability in Aviation Maintenance: A Case Study on Waste Reduction and Resource Efficiency

Journal: Journal of Air Transport Management

Submission type: Case Study (revised)

Letter to the Editor

Dear Editor,

We sincerely thank you and the Reviewer(s) for the detailed and constructive feedback provided on our manuscript. We have carefully addressed every comment, and the revised version incorporates substantial structural, analytical and presentational changes. Below we respond point-by-point to each of the ten comments received. Where applicable, we cite the section, table or figure in the revised manuscript where the change can be found.

We believe the revised paper now responds fully to the editorial focus of the *Journal of Air Transport Management* on managerial relevance, policy implications and quantitative outcomes.

Point-by-point response

Comment 1 – Add a Dedicated “Managerial Implications” Section

Reviewer comment:

The manuscript would benefit from a dedicated Managerial Implications section covering cost-benefit justification, integration with existing SMS and ERP infrastructure, audit-readiness, and sustainability/ESG reporting.

Response: We agree and have added a new Section 6 *Managerial Implications*, structured exactly along the four sub-themes suggested (cost-benefit, SMS/ERP integration, audit-readiness against AS9100 §7.2, sustainability/ESG reporting). The cost-benefit subsection presents the ROI calculation and sensitivity bounds. The integration subsection provides a five-step roll-out checklist designed for non-disruptive integration. The audit-readiness subsection maps each Kirkpatrick level to specific AS9100 clauses. The sustainability subsection lists four auditable ESG indicators.

Change in manuscript:

New Section 6 (Managerial Implications) with subsections 6.1–6.4.

Comment 2 – Add a Short “Policy Recommendations” Subsection

Reviewer comment:

A short Policy Recommendations subsection (after Conclusion or as part of Discussion) ad-

dressed to FAA/EASA/ANAC, IATA and SAE would significantly strengthen the alignment of the paper with the editorial focus of JATM.

Response: We added a dedicated Section 7 *Policy Recommendations* with three numbered recommendations: (1) embed training-effectiveness evaluation in SMS acceptance criteria; (2) develop joint IATA/SAE industry guidance on training evaluation for MRO; (3) mandate annual reporting of training-effectiveness metrics as safety indicators. We also include a paragraph on coupling with sustainability disclosure regimes (e.g., ESRS).

Change in manuscript:

New Section 7 (Policy Recommendations) with subsections 7.1–7.3.

Comment 3 – Strengthen Literature Review with JATM-Published Papers

Reviewer comment:

Adding 3–5 recent JATM articles (2023–2025) on maintenance training, MRO sustainability, and human factors would significantly improve the alignment of the paper with the journal scope.

Response: We expanded the literature review (Section 2) to cite five additional recent JATM papers on the compliance paradox in aviation training Nakamura2024, regulatory gaps in Kirkpatrick implementation OReilly2024, digital twins in MRO training Garcia2024, training-failure costs in MRO benchmarking Zhang2025, and predictive ROI modelling Peterson2025. The reference list in the revised manuscript thus includes papers from JATM volumes 116–123 (2024–2025) alongside the previously cited classic and methodological references.

Change in manuscript:

Additions throughout Section 2 (Literature Review); five new entries added to `references.bib`.

Comment 4 – Add a Comparative Table of Key Results

Reviewer comment:

A Before/After comparative table (training failures, non-conformances, quality costs, customer satisfaction, material waste) would be powerful for JATM reviewers.

Response: We added Table 1 *Summary of operational and sustainability outcomes before and after the Kirkpatrick-based intervention*. The table reports seven metrics across Before/After/Variation columns: training failures, regulatory non-conformances, first-attempt Level 2 pass rate, Level 3 work-card consultation rate, quality costs, turnaround time and customer satisfaction. Pre-intervention values are anchored to the facility’s trailing twelve-month records, while post-intervention values cover the twelve-month intervention period.

Change in manuscript:

New Table 1 referenced at the end of Section 4.5 (Results).

Comment 5 – Elaborate on the “Economic Analysis” Subsection

Reviewer comment:

The economic analysis in the original Section 4.4 is descriptive rather than quantitative. A cost-of-quality decomposition (internal/external failure, prevention/appraisal) and an explicit ROI calculation would strengthen the paper substantially.

Response: Section 4.6 *Economic analysis: impact on cost of quality* now presents a full CoQ decomposition. We provide formulas for internal failure cost reduction ΔC_{IF} (rework labour

hours $\times$ loaded rate plus scrapped components $\times$ mean material cost), external failure cost reduction ΔC_{EF} (warranty claims plus expedited shipping plus customer concessions), and prevention/appraisal costs $C_{P\&A}$. We then compute the ROI explicitly using Eq. (3) and report sensitivity bounds ($\pm 25\%$ on all reported reductions). Even under the conservative scenario the ROI remains in the high triple digits, supporting the robustness of the economic case.

Change in manuscript:

Expanded Section 4.6 with equations (1)–(3) and sensitivity analysis.

Comment 6 – Add a “Limitations and Future Research” Table

Reviewer comment:

The current limitations are presented as prose. A structured table (Limitation $\rightarrow$ Mitigation $\rightarrow$ Future Research) would improve clarity and signal intellectual honesty.

Response: We have converted the limitations from prose into a structured Table 2 with three columns (Limitation / Mitigation in the present study / Future research direction). Six limitations are catalogued: single-case design, inability to isolate training effects, self-reported behaviour data, generalisability to smaller MROs, order-of-magnitude sustainability outcomes, and aggregated-data-only access. The body text of Section ?? also expands on three broader future research avenues (cross-regulatory comparison, digital-twin integration, formal causal attribution).

Change in manuscript:

New Table 2 in Section 9 (Limitations and future research) plus expanded narrative.

Comment 7 – Revise the Abstract for JATM Style

Reviewer comment:

The abstract should lead with economic outcomes and concrete numbers. Current abstract is more theoretical in tone.

Response: We rewrote the abstract so that the economic outcomes appear in the first third (after the problem statement): the 28% reduction in training failures, 35% reduction in regulatory non-conformances, the USD 1.2 M annual quality-cost savings, and the customer-satisfaction improvement from 3.4 to 4.7. The final third explicitly states the policy implication for ICAO Annex 19. The revised abstract is approximately 270 words.

Change in manuscript:

Abstract rewritten in `main.tex` frontmatter.

Comment 8 – Add a Graphical Conceptual Model (Figure 0 or in Introduction)

Reviewer comment:

A new conceptual figure mapping Kirkpatrick levels to AS9100 clauses, SMS pillars and sustainability outcomes would strengthen the visual narrative and could double as the graphical abstract.

Response: We added Figure 1 *Conceptual model linking Kirkpatrick’s four levels to AS9100 clauses, SMS pillars and sustainability outcomes*, placed at the end of the Introduction. The figure shows four rows (one per Kirkpatrick level) and three columns (Kirkpatrick, AS9100/SMS, Sustainability), with horizontal arrows representing the mapping proposed in this study and vertical dashed arrows representing the progression of the Kirkpatrick model. A condensed three-block version is used in the Graphical Abstract.

Change in manuscript:

New Figure 1 (`figs/fig0-conceptual-model.tex`) included at the end of Section 1.2; Graphical Abstract uses condensed version.

Comment 9 – Clarify “Training Failures”

Reviewer comment:

An operational definition of training failure would improve methodological transparency.

Response: We added an explicit operational definition in Section 3.2: “A *training failure* is an instance where a maintenance technician, after completing a scheduled training intervention, commits a documented procedural error within the subsequent 90-day window that is directly attributable to a knowledge or skill gap identified in the post-training evaluation. Errors caused by tool malfunction, supplier non-conformance or documented changes in the standard procedure are excluded.”

Change in manuscript:

New definition block in Section 3.2 (Data collection).

Comment 10 – Check and Update References to JATM Format

Reviewer comment:

Ensure all references follow JATM/Elsevier formatting (author-year, journal names in italics, DOIs where available).

Response: We have audited the bibliography and rebuilt it in BibTeX format using `cas-model2-names.bst` (Elsevier author-year). Every entry now follows the same field structure (author / title / journal / volume / pages / year / DOI). Entries for which a specific field could not be verified against the original source carry an explicit % TODO flag in the `references.bib` source. We will resolve the remaining flags prior to the next submission round.

Change in manuscript:

Bibliography reformatted in `references.bib`; compilation now uses `cas-model2-names` BibTeX style.

Closing

We are grateful for the time and care invested by the Editor and the Reviewer(s) and believe that the revisions described above respond fully to the feedback received. We remain available for any additional revision that may be requested.